

Difference Between JPA, Hibernate, and Spring Data JPA

JPA (Java Persistence API)

JPA is a Java specification that provides a standard approach for mapping Java objects to relational database tables. It defines annotations such as **@Entity**, **@Table**, **@Id**, **@GeneratedValue**, **@Column**, **@OneToMany**, and **@ManyToOne**. Since JPA is only a specification, it requires an implementation such as Hibernate to execute persistence operations.

Example:

```
@Entity

@Table(name="students")

public class Student {

    @Id

    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long studentId;

    private String studentName;

    private String department;

}
```

Hibernate

Hibernate is a powerful ORM framework that implements the JPA specification. It maps Java objects to database tables, generates SQL automatically, manages caching, lazy loading, and transactions.

Example:

```
Student student = new Student();
student.setStudentName("Rahul");
student.setDepartment("CSE");
session.persist(student);
```

Spring Data JPA

Spring Data JPA is built on top of JPA and minimizes boilerplate code by providing repository interfaces such as **JpaRepository** and **CrudRepository**. It also supports pagination, sorting, and custom query methods.

Example:

```
public interface StudentRepository extends JpaRepository<Student, Long>
{
    studentRepository.save(student);

    studentRepository.findAll();
}
```

```
studentRepository.findById(1L);
```

Relationship

JPA defines the standard for persistence, Hibernate implements the standard, and Spring Data JPA simplifies data-access development by providing ready-made repository methods.

In Simple Terms: JPA – Defines the rules. Hibernate – Implements the rules. Spring Data JPA – Makes database operations faster and easier with less code.

Conclusion

JPA provides the API specification, Hibernate performs the actual ORM operations, and Spring Data JPA further improves productivity with built-in CRUD methods, pagination, sorting, and query generation.